Shaojie Chen

305-310-5596 | cheerioov2@gmail.com | Clinton Township, MI

MOTORSPORTS AERODYNAMICS ENGINEER

SUMMARY

Motorsports-focused Mechanical Engineer with an M.S. and hands-on experience in vehicle aerodynamics, CFD/CAD development, active aero, embedded controls, CAN data acquisition, native iOS HMI requirements, and track-test correlation. Developed a C7 Corvette active-aero ecosystem spanning aerodynamic surfaces, ESP32 actuator control, position feedback, Auto/Manual mode logic, mobile telemetry/debug visibility, Pi Toolbox analysis, and repeatable road/track validation. Proficient with STAR-CCM+, ANSYS Fluent, NX, SolidWorks, Python, MATLAB, and multidisciplinary test-data interpretation; graduate thesis focused on CFD.

MOTORSPORTS AERODYNAMICS & VEHICLE SYSTEMS

C7 Corvette Active Aero / Vehicle Performance Development Platform **Sep 2024 - Present**

- **Aerodynamic Design / CFD:** Designed and iterated front and rear aerodynamic surfaces and components in CAD; used STAR-CCM+ and ANSYS Fluent CFD studies to evaluate drag, downforce, aerodynamic balance, packaging, and stability tradeoffs across operating states.
- **Embedded Controls / CAN:** Developed ESP32-based actuator control with real-time CAN decoding for throttle, brake, speed, and steering-wheel inputs; implemented Auto/Manual deployment logic, position feedback, manual override, safety handling, and Wi-Fi monitoring for front and rear aero hardware.
- **Native iOS HMI / SwiftUI:** Directed native iOS HMI development in SwiftUI/Xcode for the active-aero system; specified physical front/rear mapping, telemetry readability, interface hierarchy, control-state visibility, and engineering-debug views.
- **Controls UX / Safety Logic:** Prevented command conflict by hiding manual controls during Auto operation, limiting linked front/rear adjustment to Manual mode, removing unnecessary Remote STOP behavior, and retaining immediate STOP for direct actuator motion.
- **Embedded-to-Mobile Integration:** Translated ESP32 firmware handoff and actuator-protocol details into app-side requirements for mode state, command mapping, position feedback, and debug visibility; aligned mobile controls with actual hardware behavior and safety constraints.
- **Track Testing / Correlation:** Conducted baseline road and Gingerman Raceway testing; analyzed suspension travel in Pi Toolbox and established repeatable test-analyze-update loops to correlate aero state, driver inputs, platform response, and vehicle balance.
- **Vehicle Instrumentation:** Developing Nitron R3 strain-gauge load measurement, suspension-travel sensing, and tire-temperature thermal imaging to quantify corner loads, platform response, and contact-patch behavior under changing aero states.
- **Vehicle-Dynamics Simulation:** Developing a Simulink/CarSim workflow to complement measured track data, explore aero and vehicle-dynamics interactions, and improve setup and control-strategy decisions before additional testing.
- **Rapid Prototyping / DFM:** Built and iterated functional hardware in-house using 3D printing and hands-on fabrication to validate fitment, kinematics, packaging, serviceability, and manufacturability; resolved DFM/DFA issues through fast design loops.

CORE MOTORSPORTS CAPABILITIES

Aerodynamics / Vehicle Performance: active aero, aero surfaces and components, drag/downforce/balance/stability tradeoffs, CFD-to-test correlation, suspension response, tire behavior, and track-focused development
CFD / CAD / Analysis: STAR-CCM+, ANSYS Fluent, NX, SolidWorks, CATIA, Creo, AutoCAD, MATLAB, Python, Pi Toolbox/Cosworth Tool, and engineering data visualization
Controls / HMI / Data: ESP32, CAN, actuator position feedback, Auto/Manual modes, SwiftUI/Xcode HMI direction, telemetry/debug UI, C#, MicroPython, Arduino, and PLC controls
Testing / Instrumentation: road/track validation, high-speed DAQ, strain gauges, suspension travel, thermal imaging, test planning, transient analysis, root-cause investigation, and technical reporting

MOTORSPORTS DEVELOPMENT APPROACH

End-to-End System Engineering: connect aerodynamic surfaces, actuation, embedded control, mobile HMI, telemetry, vehicle signals, and track-test workflows into one integrated development platform
Correlation Workflow: combine CFD results, suspension travel, CAN data, actuator position, tire temperature, and Simulink/CarSim models to guide iterative hardware and control changes
Engineering Communication: translate firmware/hardware handoffs, test findings, and integration issues into prioritized requirements, debug plans, and cross-functional recommendations

TRACK EXPERIENCE & PERFORMANCE DATA

- Sebring International Raceway: 2:27 personal best - stock 2017 Camaro SS 1LE, Hawk DTC60, Goodyear Supercar 3.
- Homestead-Miami Raceway: 1:39 personal best - stock 2017 Camaro SS 1LE, Hawk DTC60, Goodyear Supercar 3.
- Gingerman Raceway: 1:40 personal best - 2019 Corvette Grand Sport, Hawk DTC60, Michelin Pilot Sport 4S; used baseline suspension-travel data for active-aero correlation.

SHAOJIE CHEN | PROFESSIONAL ENGINEERING EXPERIENCE

Enginuity Power Systems **Detroit, MI**
Senior Project Engineer ***June 2024 - Present***

- Developed and validated a custom long-reach spark plug solution for a high-efficiency opposed-piston engine, resolving misfire and high-RPM issues through iterative testing, supplier coordination, and material validation.
- Evaluated hybrid inverter and battery behavior with high-speed DAQ and power analysis, characterizing waveform quality, transfer timing, efficiency, current ripple, and transient operating behavior to guide system decisions.
- Lead multi-subsystem design and controls integration for a hybrid energy platform coordinating generator, inverter, battery, sensors, and thermal loads; define control logic, protection strategies, and validation plans across hardware and software interfaces.
- Develop electrical and thermal load-emulation capability to create repeatable test conditions for controls calibration, transient testing, troubleshooting, and system-level performance validation.
- Create and review CAD layouts, prototype concepts, brackets, test fixtures, and assembly interfaces; support hands-on builds using 3D-printed parts, sensors, wiring, and light fabrication.
- Support mechanical and electrical architecture definition, including controller I/O, enclosure ventilation, sensor integration, protection concepts, component selection, packaging, and serviceability requirements.
- Direct international supplier collaboration and root-cause investigations across electromechanical and power-electronics subsystems, coordinating firmware changes, protection improvements, specification alignment, and hardware validation.
- Translate test data, engineering constraints, and supplier findings into technical documentation, design recommendations, and management-ready presentations for cross-functional decisions.
- Develop commissioning and remote-connectivity documentation for ComAp-based CHP systems, covering 4G/LAN setup, remote diagnostics, firmware/archive updates, and field handover requirements.
- Directed engineering development and supplier collaboration for portable power stations and solar modules, leading firmware coordination, root-cause analysis, and protection improvements including low-voltage cutoff and current limiting.

Development Engineer ***April 2023 - June 2024***

- Executed R&D test plans for the 8 kW micro-CHP platform, built data-driven validation routines, and accelerated project timelines.
- Developed programmable electrical load-bank automation that improved test accuracy, repeatability, and performance-validation capability in the test cell.
- Enhanced embedded controls using CLICK PLC and Arduino/ESP32 platforms, implementing safety interlocks, limit logic, auxiliary device control, and cleaner code structures for safer operation.
- Established an in-house additive-manufacturing workflow, trained technicians, and reduced prototype cycle time by approximately 80%.
- Coordinated testing and technical specification alignment with international generator and inverter manufacturers to support system integration, efficiency evaluation, and validation of electrical interfaces.

Patent Pending: METHODS AND DEVICES FOR SELECTING A POWER SOURCE

EDUCATION

University of Miami **Miami, FL**
Master of Science in Mechanical Engineering ***Dec 2022***
Thesis: An Investigation of the New High-Efficiency Three-Cylinder Internal Combustion Engine (CFD Focused)
Bachelor of Science in Mechanical Engineering ***May 2021***
Capstone Design: Compressed Air Supercharging of an Internal Combustion Engine
Relevant Coursework: Fluid Analysis, System Dynamics, Heat Transfer, Thermodynamics, Finite Element Method, Additive Manufacturing

ADDITIONAL TECHNICAL TOOLKIT

Engineering / Analysis: Abaqus, finite-element methods, CAD-driven product development, manufacturing drawings, DFM/DFA, technical documentation, and cross-functional design reviews
Controls / Instrumentation: Dewesoft DAQ, differential voltage probes, current clamps, strain gauges, thermal imaging, CAN, Modbus/RS-485, CLICK PLC, ESP32/Arduino, and actuator feedback
Programming / Communication: Python, MATLAB, C#, MicroPython, SwiftUI/Xcode development direction, supplier coordination, test reports, technical presentations; Languages: Chinese (Native), English (Advanced)

SELECTED TRANSFERABLE ENGINEERING STRENGTHS

Systems Integration: connect mechanical hardware, electronics, controls, sensors, software interfaces, safety logic, and serviceability requirements into practical development platforms
Test Engineering: define validation plans, select instrumentation, diagnose measurement limitations, analyze transient data, correlate results, and convert findings into design changes
Product Development: move concepts through CAD, prototype build, embedded controls, bench validation, supplier review, DFM/DFA, documentation, and field-readiness decisions
Technical Leadership: coordinate international suppliers and cross-functional teams, translate complex issues into prioritized requirements, and communicate recommendations to management